DATA ANALYTICS PROJECT

FRAUD
TRANSACTION ANALYSIS

Prepared by: **Rahul** | Tools: **Python · MySQL · Power BI**

Dataset: Fraud_Transaction_Analysis_Rawdataset.csv | Date: March 2026

Project Overview

OBJECTIVE

Perform a comprehensive end-to-end analysis of 10,000 fraud transactions to uncover hidden patterns, identify high-risk segments, and deliver actionable insights to help financial institutions
detect and prevent fraud effectively.

- Identify overall fraud rate across all transactions
- Discover high-risk types, categories & countries
- Analyse time-based fraud patterns & peak hours
- Evaluate device & IP risk scores as predictors
- Build interactive Power BI dashboard

TOOLS USED

Microsoft Excel
Initial data viewing & pivot analysis

Python / Pandas
Cleaning, EDA, feature engineering

MySQL
Data storage & SQL business analysis

SQLAlchemy
Python → MySQL integration

Power BI
Interactive dashboard & visuals

// 02 DATASET SUMMARY

Dataset Summary

10,000	10	500	9,500	5.0%
TOTAL RECORDS	COLUMNS	FRAUD CASES	LEGITIMATE	FRAUD RATE

KEY COLUMNS

COLUMN	DATA TYPE	DESCRIPTION
transaction_id	int64	Unique identifier per transaction
amount	float64	Transaction amount in USD
transaction_type	object→category	ATM / POS / Online / QR
merchant_category	object→category	Food, Travel, Electronics, Clothing, Grocery
country	object→category	US, UK, FR, DE, TR, NG
hour	int64	Hour of day (0–23)
device_risk_score	float64	Device risk score (0.0 – 1.0)
ip_risk_score	float64	IP address risk score (0.0 – 1.0)
is_fraud	int64	TARGET: 1 = Fraud, 0 = Legitimate

Data Cleaning & EDA

01 Data Loading

pd.read_csv() loaded 10,000 rows × 10 columns. df.head(10) previewed structure.

02 Inspection

df.shape confirmed (10000, 10). df.dtypes revealed object columns needing conversion.

03 Null & Duplicate Check

df.isnull().sum() = 0 nulls across all columns. df.drop_duplicates() — no duplicates found.

04 Type Standardization

transaction_type, merchant_category, country → astype('category'). is_fraud, hour → int64.

05 Feature Engineering

4 new columns: is_high_amount, is_night_txn, is_high_risk, amount_bucket (pd.cut).

06 MySQL Export

df.to_sql() with SQLAlchemy loaded all 10,000 rows into fraud_db.fraud_transactions.

Python → MySQL

Clean CSV
fraud_cleaned.csv

→

SQLAlchemy
create_engine()

→

fraud_db
MySQL database

→

fraud_transactions
10,000 rows loaded

CONNECTION PARAMETERS

Username	root
Host	localhost
Port	3306
Database	fraud_db
Table	fraud_transactions
Rows Loaded	10,000
Method	df.to_sql()
if_exists	replace

FEATURE ENGINEERING RESULTS

is_high_amount
amount > 1000 → 1 else 0

is_night_txn
hour < 6 OR hour >= 22 → 1

is_high_risk
device > 0.7 AND ip > 0.7 → 1

amount_bucket
pd.cut() → 5 price ranges

SQL Business Analysis

Q1 Overall Fraud Rate

10,000 total → 500 fraud → 5.00%

💡 *5% of all transactions are fraudulent.*

Q2 Amount: Fraud vs Legit

Fraud avg $1,657.58 | Legit avg $100.28

💡 *Fraud is 16x higher in value than legit.*

Q3 Peak Fraud Hours

Hour 4 = 100% fraud rate (117 cases)

💡 *4 AM is a critical fraud alert window.*

Q4 Night vs Day Fraud

Night: 17.92% | Day: 3.19% fraud rate

💡 *Night fraud is 5.6x higher than daytime.*

Q5 Fraud by Country

NG: 100% | US: 4.73% | UK: 4.33%

💡 *Nigeria = 100% — every transaction is fraud.*

SQL Business Analysis

Q6 Fraud by Transaction Type

ATM: 5.46% | Online: 5.26% | QR: 4.79% | POS: 4.52%

💡 *ATM has highest fraud count (138) and rate.*

Q7 Fraud by Merchant Category

Clothing: 5.50% | Travel: 5.26% | Electronics: 4.88%

💡 *Clothing & Travel need extra verification.*

Q8 Risk Score Effectiveness

Fraud: device 0.858 / IP 0.852 vs Legit: 0.148 / 0.150

💡 *Risk scores are 5.8x higher for fraud.*

Q9 Multi-Factor High-Risk Profile

Risk > 0.7 + Amount > $1,000 = 100% fraud rate

💡 *All multi-factor high-risk combos are fraud.*

10 Critical Findings

01 **5% overall fraud rate**
500 fraud out of 10,000 transactions

02 **Fraud amounts 16x higher**
Avg $1,657.58 fraud vs $100.28 legit

03 **4 AM = 100% fraud hour**
All 117 transactions at this hour are fraud

04 **Night fraud 5.6x higher**
17.92% night vs 3.19% daytime rate

05 **Nigeria = 100% fraud**
Every NG transaction is fraudulent

06 **ATM most vulnerable**
5.46% rate — highest of all channels

07 **Clothing most targeted**
5.50% rate — highest merchant category

08 **Risk scores highly predictive**
0.858 fraud vs 0.148 legit device risk

09 **$1K–5K = 51% of fraud**
255 out of 500 fraud in this range

10 **Multi-factor = 100% fraud**
High risk + high amount always fraud

Interactive Dashboard

5.0%	$1,658	500	9,500	04:00
Fraud Rate	**Avg Fraud Amt**	**Total Fraud**	**Legitimate**	**Peak Hour**
500 of 10,000	vs $100 legit	Confirmed cases	95% of total	117 detected

DASHBOARD VISUALS

KPI Cards ×4	Fraud rate, avg amount, total fraud, legit count
Line Chart	Fraud count by hour — 4 AM spike visible
Bar Chart	Fraud by merchant category — Clothing leads
Clustered Bar	Transactions by channel — ATM leads fraud
Bar Chart	Fraud by country — NG = 100% fraud rate
Donut Chart	Fraud by amount bucket — $1K–5K = 51%
Gauge Visual	Avg device risk — 0.858 for fraud cases

DAX MEASURES

Fraud Rate %
```
DIVIDE(SUM(is_fraud), COUNTROWS(...))*100
```

Total Fraud Count
```
CALCULATE(COUNTROWS(...), is_fraud=1)
```

Avg Fraud Amount
```
CALCULATE(AVERAGE(amount), is_fraud=1)
```

Legit Count
```
CALCULATE(COUNTROWS(...), is_fraud=0)
```

Peak Fraud Hour
```
VAR-based formula → returns '04:00'
```

Avg Device Risk
```
CALCULATE(AVERAGE(device_risk), fraud=1)
```

Business Recommendations

IMMEDIATE ACTIONS

→ Block/flag all Nigeria (NG) transactions — 100% fraud rate

→ Enable 2FA for all transactions between 10 PM and 6 AM

→ Deploy real-time alert system for 4 AM window transactions

RISK SCORE THRESHOLDS

→ Auto-flag when device_risk > 0.7 AND ip_risk > 0.7

→ Add secondary verification when either score > 0.5

→ Integrate risk scoring into real-time transaction pipeline

TRANSACTION MONITORING

→ Add extra verification for ATM — highest fraud rate (5.46%)

→ Flag all transactions exceeding $1,000 — 51% of fraud

→ Monitor Clothing & Travel categories — highest fraud rates

LONG-TERM PREVENTION

→ Build ML model using risk scores, amount, hour, country

→ Implement customer loyalty programs for anomaly detection

→ Conduct quarterly geographic risk reviews & update rules

Conclusion

This project demonstrated a complete data analytics pipeline — from raw CSV to actionable business insights — using Python, MySQL, and Power BI. Analysis of 10,000 transactions revealed a 5% fraud rate with clear, exploitable patterns.

Excel	Python	MySQL	SQL	Power BI
Pivot tables, column review	Clean dataset + 4 new features	10,000 rows in fraud_db	9 business questions answered	Interactive dashboard built

By implementing recommended risk score thresholds, geographic controls, and time-based verification — financial institutions can significantly reduce fraud exposure without impacting legitimate customer experience.